

1-C:

A – When private benefits exceed private costs, this represents “net-private benefits”

B – Private benefits cannot exceed social benefits.

C – When social benefits exceed private benefits, this represents the existence of “positive externalities”, since $SP - PB = EB$. This is why, this becomes correct.

D – When social benefits exceed private costs, this does not represent “external benefits” either.

2-B

A – When private benefits are greater than private costs, the private sector, who is only interested in private costs and benefits, will approve of the project.

B – When social benefits are greater than social costs, this represents the existence of “net social benefits”, and this implies that this project will be approved by the government because **“it is socially worthwhile”**

C – Social benefits should not be greater than total costs(that represent the private costs of production), but instead , they should be greater than social costs, in order for the project to be approved by the government.

D – When total costs(private costs) are greater than the total revenue(private benefit to a producer), the private sector would not even approve of it.

3-B:

Allocative efficiency refers to the optimal allocation of resources in an economy, where resources are allocated to maximize the satisfaction of society as a whole.

Income inequalities (B) do not directly pose a threat to allocative efficiency, as they relate to the distribution of income among individuals and is a macro-economic concern, not related to the allocation of resources.

The other options do pose a threat to allocative efficiency:

A) Imperfect information (A) can lead to consumers making suboptimal choices, reducing allocative efficiency.

C) Externalities (C) can cause market failures. There is market failure associated with under-allocation of merit goods (goods with positive externalities) and over-allocation of de-merit goods (goods with negative externalities).

D) Monopolistic elements (D) can lead to market failure associated with monopoly exploitation.

So, income inequalities (B) are the correct answer, as they do not directly affect the allocation of resources.

4-D:

A – T to P does not represent a giffen good; giffen good is identified through the income effect from U-P, which outweighs the substitution effect. We cannot say if T to Q represents perfect elasticity because we do not have information about price-elasticity of demand.

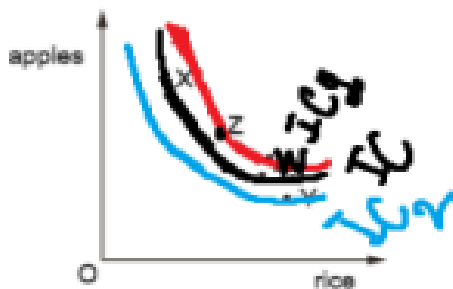
B- T to R represents the total effect from which **"T to U" that shows a movement along the indifference curve** represents the "substitution effect", and therefore, **U to R** becomes the "income effect"

C - **"T to U" that shows a movement along the indifference curve** represents the "substitution effect", but **"U to S"** shows the income effect of a "normal good", which helps us to identify that a particular good is a normal good.

D - **"T to U" that shows a movement along the indifference curve** represents the "substitution effect", and **U to P** represents the income effect of the **"giffen good"**, since in the case of giffen goods, income effect outweighs the substitution effect.

This is why, this becomes correct.

5-D:



A – The consumer is not "indifferent" between combination Y and Z because combination Z is on a higher indifference curve which generates a greater level of satisfaction as compared to combination Y.

B - The consumer is not indifferent between combinations Z and W , this is because, combination Z is on a higher indifference curve as compared to combination W. A rational consumer is only indifferent along the same indifferent curve.

C – The consumer prefers combination X to combination Y since "X" is on a higher indifference curve than "Y"

D – This is correct as the consumer would prefer combination Z to combination W because "Z" is on a higher indifference curve, and thus yields a greater level of satisfaction.

6-D:

The schedule shows the short-run marginal cost of producing good X.

$FC = 50$

units of X	0	1	2	3	4	5
marginal cost (\$)		45	40	30	20	20

Given that the total fixed cost is \$50, which level of output minimises average total cost?

- A 2 units B 3 units C 4 units D 5 units

	TC	ATC
* output level		
1	95	95
2	135	67.5
3	165	55
4	185	46.25
5	205	41

Handwritten notes:

- $TC = FC + VC$ OR $TC \rightarrow$
- * TC at the previous level of output + marginal cost
- * $ATC = TC / Qty$
- lowest average total cost.

7-A:

- ➔ Price discrimination refers to the practice of charging different prices to different customers or in different markets for the same product or service. This means that the reasons are not associated with cost, but instead depend upon price elasticity of demand.

Option A: Different price elasticities of demand. This option is correct because for price discrimination to be profitable, there must be varying degrees of price sensitivity among customers in different markets. If the price elasticity of demand differs across markets, the firm can charge higher prices to customers with relatively inelastic demand (less price-sensitive) and lower prices to customers with relatively elastic demand (more price-sensitive). By doing so, the firm can extract more revenue from customers who are willing to pay higher prices while still capturing customers with more price-sensitive demands. This is why, key A, becomes correct.

Option B: Price discrimination focuses on capturing different levels of willingness to pay among customers, which is determined by their price elasticity of demand, and not "price-elasticity of supply"

Option C: It can be practiced by a single firm (even a monopoly) that offers the same product or service, but charges different prices to different customer segments or markets.

Option D: Different products. This option is incorrect. Price discrimination is applied to the **same product or service**, offering different prices based on customer characteristics or market segmentation.

8-C:

A – This is incorrect because Average fixed cost can not be zero, and this is the reason why, the average fixed cost curve is shown through a curve rather than a linear line.

B – Average revenue is equal to Average cost when “normal profits” are made.

C – In-order to maximize total revenue, **the marginal revenue must be “zero”**. Here, we are also interested in parallelly achieving the objective of “profit-maximization” too. So now if, the **Marginal cost(MC)= zero**, and **Marginal revenue(MR) is also equal to zero**(in the pursuit of the objective of revenue maximization), we will be able to maximize profits too, since the condition of **MC=MR** is fulfilled.

D – When marginal revenue is greater than marginal cost, the total output needs to be increased such that : **MC=MR**, in order to maximize profits.

9-B:

As a result of technological change when the industry grows, and now the **same level of output can be produced at a lower cost**, this means that **more firms will be encouraged to enter, and now the number of firms in the industry will rise**.

The minimum efficient scale (MES) point refers to the level of production or output at which a firm can achieve the lowest possible average cost per unit. It represents the optimal scale of operation for a company to maximize efficiency and minimize costs.

At the MES point, a firm has reached the minimum level of production necessary to fully utilize its available resources and achieve economies of scale

And since, the MES point has reduced, this means that now **at a “lower level of output”, firms can minimize their long-run average total costs, and exploit the maximum economies of scale, as beyond the MES point, no further economies of scale can be exploited**.

“Therefore, firms are likely to produce at a lower level of output, at which MES is achieved, as growing beyond this size would lead to them experiencing **dis-economies of scale**” that are cost-disadvantages in form of a rise in long-run average costs as a result of them growing large in size. This suggests that there will be a **reduced size of the firms, as they are more-likely to operate on a smaller scale**.

10-D:

A: Here, the profit is maximized because equilibrium is achieved where **MC=MR**

B: Here, the sales revenue is maximized because the marginal revenue is equal to zero

C: Here, allocative efficiency is achieved because **marginal cost= Average revenue (price)**

D : This is correct because this shows that ‘normal profits’ are being made. This is because, Average revenue is equal to Average cost.

11-A:

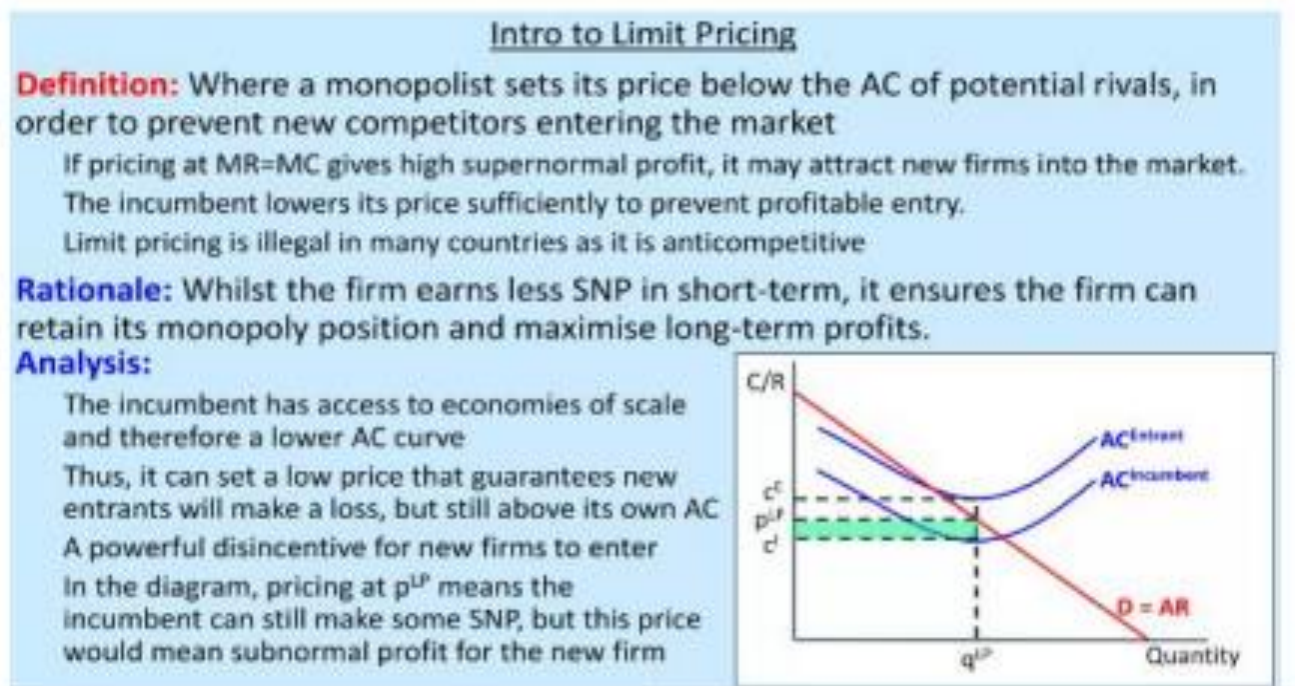
A: If there are economies of scale in certain industries, this means that the firms operating on a larger scale will be having cost-advantages, as a result of which, the firms operating on a smaller scale wouldn't be able to survive.

B: Small firms often supply personalized or customized services to consumers, as a result of which they can charge higher prices from consumers. This does not hinder their survival

C: Small firms often target a niche (limited number of customers), and this does not hinder their survival either. If a small firm engages itself in selling luxury perfumes to a niche market, this means that it can also charge higher prices, making survival easier

D: Small-owner managed firms have less risk of perhaps dis-economies of scale, which is likely to increase the chances of survival

12-D:



A – Although, “limit pricing” strategies are meant to discourage potential rivals from entering the industry, it cannot certainly restrict the industry “to a target number of firms”. This is why, this becomes incorrect.

B – Limit pricing is not aimed to eliminate existing rival oligopolists, instead it is meant to discourage or prevent new rivals from entering the industry.

C – By making use of limit pricing policy, the oligopolist will not make a “satisficing level of profit”. As can be seen in the diagram above, it will still make “super-normal profits”.

D – This is correct because, initially low prices may lead to the firm sacrificing on “short-term profits”, but after the potential threat of competition is eliminated, firms can raise their prices again, which will “lead to a long-term profit”.

13-A:

A- A ban will be the most effective way of reducing road traffic congestion arising from usage of private cars, **considering the fact that the demand for private cars is “highly inelastic”**. This is why, key A, will be correct.

B and D: As a result of introducing subsidies to lower the prices of using bicycles(option B) and public transport, such as trains and buses(option D), the qty demanded of private cars would not contract by more, as very few people will switch to cycles or public transport, which means that the policy is **ineffective**.

C : Even if prices of using private cars has increased, due to a policy of “road pricing”, qty demanded would contract by a relatively smaller percentage, as very few people will switch to cycles or public transport, which means that the policy is ineffective.

14-C:

Option A: A decline in the share of wages in national output would actually suggest an increase in wage inequality, not a reduction. This is because a smaller share of national output going to wages implies a larger share going to capital, which tends to benefit higher-income individuals.

Option B: A decline in trade union bargaining power would likely lead to an increase in wage inequality, not a reduction. Trade unions tend to push for higher wages and better working conditions, which benefits lower- and middle-income workers. With reduced bargaining power, workers may struggle to negotiate better wages, leading to increased inequality.

Option C: The existence of a minimum wage can help explain a reduction in wage inequality during the post-2008 economic recession. A minimum wage sets a floor for wages, ensuring that low-income workers earn a certain level of income. During a recession, when wages might otherwise fall, a minimum wage can help maintain a more equal distribution of income. This is because the minimum wage protects low-income workers from further wage declines, reducing the gap between high- and low-income earners. Therefore, option C is the correct answer.

Option D: A trend towards later retirement might have some indirect effects on wage inequality, but it is not a direct cause of reduced wage inequality. Later retirement could lead to a more experienced workforce, potentially increasing productivity and wages.

15-A:

Marginal Revenue Product (MRP)

MRP measures the additional revenue generated by a unit increase in the quantity of a variable input (such as labor). It's calculated by multiplying the:

1. **Marginal Physical Product (MPP)**: The additional output generated by a unit increase in the variable input.
2. **Marginal Revenue (MR)**: The additional revenue generated by a unit increase in output.

$$\text{MRP} = \text{MPP} \times \text{MR}$$

16-A:

This explains that key A will be correct.

16 A firm currently employs 30 workers at a daily wage of \$50 each.

The marginal cost of employing one extra worker is \$112 per day.

By how much will the firm have to increase the daily wage in order to increase its labour force from 30 to 31 workers?

- A \$2 B \$4 C \$62 D \$112

Change in Total cost arising from employing an additional unit of labor.

$$MC_{11} = TC_{11} - TC_{10}$$

① $112 = [x \times 31] - [50 \times 30]$ (Note: x is the new wage rate)

② $112 = 31x - 1500$

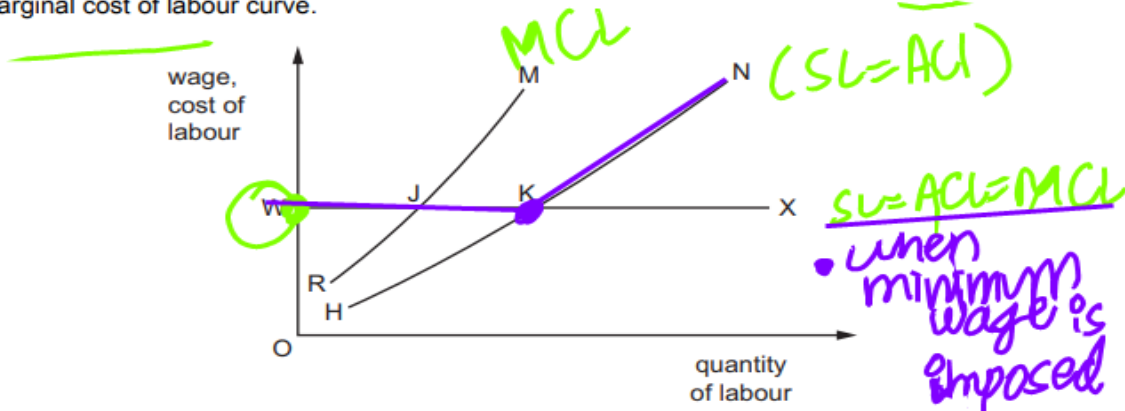
③ $\frac{112 + 1500}{31} = x \Rightarrow x = 52$

Note: Let x represent the new wage rate.

① previous wage $\rightarrow 50$
② new wage $\rightarrow 52$
③ The wage rate rose by \$2

17-D:

In the diagram, HN is the initial supply of labour curve faced by a firm, and RM is its initial marginal cost of labour curve.



What will be the firm's new labour supply curve if the workers join a trade union and achieve a union negotiated wage, OW?

- A RJX B HKX C WJM **D WKN**

18-B:

- Output per worker represents “**labor productivity**”, which is also known as the “Average product of labor”
- We are interested in an option in the case of which, the industry’s total product(output) is greater than the Average product of labor.

The explanation below explains why key B will be correct.

labor productivity
Output per worker in an industry increases more slowly than the industry's total output.
What could explain this?

labor productivity = $\frac{\text{Total Output}}{\text{No. of workers / Employment}}$

Numerator ↑
But denominator is also rising

Total Output ↑
some of the gains arising from an increase in output will be offset because the number of workers employed are also rising. **therefore output per worker will increase slowly.**

labor productivity

A - There will not be a decrease in labor productivity; this is because, the question says that output per worker is **increasing slowly**.

C - Again, an increase in overtime working by the existing workforce will result in a rise in output per worker.

D – An increase in the hourly wage rate when increases the motivation of workers, results in a rise in the Average product of labor.

19-D:

A : A change in the slope of the **ppc** from concave to convex represents that the opportunity cost changes from increasing to decreasing.

B : A movement from actual(point inside the ppc) to potential(point on the ppc) represents short-run economic growth or actual economic growth.

C : Movement from one point of the PPC to the other represents the concept of “opportunity cost”

D : An outward shift of the ppc represents long-run, potential economic growth. This represents that the long-run Aggregate supply of the economy is increasing. As a result of which, key D, becomes correct.

20-D:

Economic Growth:

- Refers to an increase in a country's production capacity, typically measured by the increase in Gross Domestic Product (GDP).

Economic Development:

- Encompasses not only economic growth but also social and structural transformations.

- Aims to improve the standard of living, well-being, and quality of life for citizens.

A - Economic growth driven by maximizing raw materials extraction is unlikely to promote economic development. This approach can lead to overexploitation and depletion of natural resources, causing environmental degradation.

B – A continuous focus on increasing output represents rapid economic growth, which may be unsustainable.

C - Discovering new resources in remote natural environments may lead to economic growth, but it is not necessarily a path to economic development.

This approach can result in environmental damage and displacement of local communities which may be detrimental for economic development.

D- Economic growth that provides funds to finance cleaner production processes is most likely to promote economic development. This approach prioritizes sustainability and environmental protection, reducing the negative impacts of economic activity

21-B:

Foreign aid refers to the international movement of money, services, or goods from governments or international institutions for the benefit of the receiving country or its citizens. **Official development assistance** (govt to govt transfer of aid) is the most common form of foreign aid, which is the help provided to support development and to fight poverty.

Foreign Direct Investment (FDI) involves foreign investors directly investing in another nation's productive assets. Conversely, Foreign Portfolio Investment (FPI) entails investing in financial assets, like stocks and bonds, of entities situated in a different country.

Both of these are classed as private investment.

This explains why, option B, is correct.

22-A:

A - Using actual market exchange rates to compare living standards between countries can be misleading, as exchange rates can fluctuate and may not accurately reflect the relative purchasing power of different currencies. **PPP rates, on the other hand, take into account the different prices of goods and services in each country, providing a more accurate comparison of living standards.**

Using market exchange rates might overvalue the GNP per head of Country X, making it appear higher than Country Y's, even if the actual purchasing power is similar or lower.

B – Transfer payments are not included in gnp

C – Inflation has already been accounted for in the real gnp/head data.

D - The distribution of income (more unequal in Country X) is important for understanding poverty and social welfare, but it doesn't affect the ranking of GNP per head, which is a measure of average income.

23-C:

* Total labour force $\Rightarrow 36\text{ m}$
* Unemployment rate $\Rightarrow 10\%$
* Total number of unemployed people $\rightarrow 10\% \times 36$
 $\hookrightarrow 3.6\text{ m}$

24-C:

A - The decline in the manufacturing share of real GDP over time suggests a structural change in the economy, with manufacturing becoming a smaller sector perhaps due to **de-industrialization as the economy develops**. This can lead to structural unemployment, as workers may be **occupationally immobile** and may not have the skills or training to adapt to new industries or jobs

A: Cyclical unemployment refers to job losses due to economic downturns or recessions over the whole economy.

B: Frictional unemployment refers to temporary unemployment arising from workers in between jobs.

D: Voluntary unemployment refers to individuals choosing not to work at the current wage rate, which is not related to the economic structural change described.

25-B:

$$\begin{aligned}
 & \begin{array}{l} \text{*(AE)} \\ * I \Rightarrow 100 \\ * \Delta NI \Rightarrow 300 \end{array} \quad \begin{array}{l} * \text{multiplier} \Rightarrow \\ \downarrow \\ \text{formula} \end{array} \Rightarrow \frac{\Delta NI}{\Delta AE} \Rightarrow \frac{300}{100} \Rightarrow \textcircled{3} \\
 & * \frac{1}{1 - MPC} \Rightarrow 3 \\
 & * \frac{1}{1 - x} \Rightarrow 3, \quad 1 = 3 - 3x \\
 & * 1 - 3 = -3x \\
 & * -2 = -3x \\
 & * \frac{-2}{-3} = x \Rightarrow \textcircled{x(MPC) \Rightarrow \frac{2}{3}}
 \end{aligned}$$

26-A:

When the central bank **buys** back government bonds from the general public or the non-bank private sector, this will increase the money supply, representing quantitative easing.

As a result of which, there will be a surplus of money in the money market, which results in **a fall in the interest rate**.

This will lead to a fall in the costs of borrowing, as a result of which, **investment demand increases**.

This will result in a rise in Aggregate demand and subsequently real GDP.

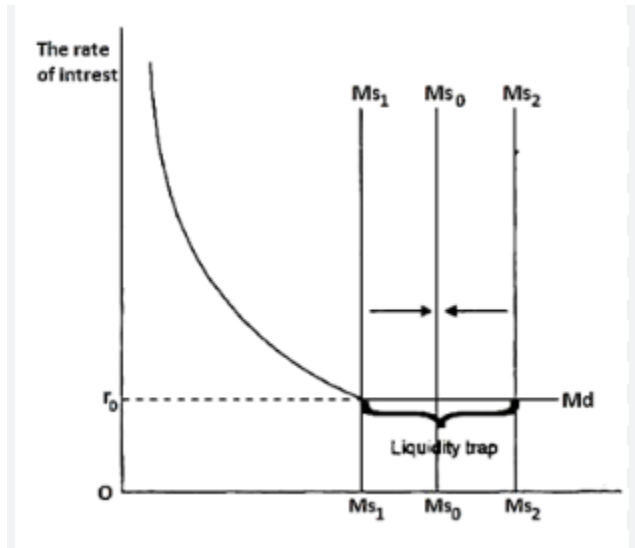
27-C:

A - If bond prices are expected to rise, this represents that there might be a decrease in interest rates.

B: If investment demand is interest-inelastic, an increase in the money supply would still lead to lower interest rates.

C - When the nominal interest rate is very close or equal to zero, the opportunity cost of holding money becomes zero, and economic agents—banks, firms, or individuals—tend to hoard money even if they have more money than they need. Most importantly, traditional monetary policy becomes ineffective in stimulating

the economy. Even when the monetary authority increases the money supply, interest remains unchanged, and so there is no expansionary effect on Aggregate demand.



D: An increase in the velocity of circulation (the rate at which money is spent and respent) would not have a direct impact on interest rate.

28-D:

An economy will move from point X to point Y when there is an increase in Aggregate supply.

A – An increase in government spending will increase Aggregate demand.

B – An increase in income tax, after having reduced disposable income and consumption expenditure, will decrease Aggregate demand.

C – Whereas, an increase in average wage rate will increase costs of production, thereby leading to a decrease in Aggregate supply.

D – An increase in labor productivity will reduce costs of production as labor costs fall, which will bring about an increase in Aggregate supply, thereby making this correct.

29-B:

A – A higher foreign exchange rate since will lead to a BOT deficit in the long-run, as it makes exported goods expensive and imported goods cheaper, will result in a fall in Aggregate demand, because of which, there will be a higher degree of unemployment.

B – A low rate of inflation will make export oriented goods relatively cheaper, thereby attracting an extension in international qty demanded by a relatively higher percentage when the demand is elastic, which will result in an increase in export revenues. These will contribute to a BOT surplus. This is why, this becomes correct.

C – An even distribution of income is not directly linked with a higher rates of savings.

D – A rapid economic growth is unlikely to be sustainable, because this involves resources being exploited unsustainably.

30-C:

A – The discovery of oil reserves in a developing economy will not lead to a more equal distribution of income, which depends on the taxation and benefits policies of the government.

B – Right now only oil reserves are being discovered, this means that is not likely to improve the competitiveness of commercial agriculture.

C – Since this will reduce oil imports by the developing economy, it will lead to a fall in supply of currency, which will lead to the exchange rate to appreciate.

D – Oil discovery is going to bring about a reduction in the volume of oil imports and not imports of oil products.

